TAIR-WM: Training-Free Watermark Detection via Exact-DDIM Inversion and Task-Aware Statistical Scoring

Working draft; designated results remain to be completed.

Abstract

Frequency-domain generative watermarks recover structured initial noise to verify image provenance and identify registered keys. Geometric transformations disturb both recovery and template matching. We present TAIR-WM, a training-free detector with three technical components. Residual-guided Exact-DDIM adapts iterative inverse-step refinement to HSQR detection, using a finite evaluation budget, a stopping tolerance, and per-image best-candidate retention. Covariance-aware verification centers correct-key feature residuals and scores them through shrinkage-covariance whitening. Task-aware dual-metric detection shares the recovered features but uses Mahalanobis scoring for verification and Complex-L1 for identification, selected by an externally supplied task. Neural weights remain fixed; residual statistics are fitted offline. Across twelve nonzero bilinear rotations, the combined verification branch improves mean TPR at FPR below 1% from 3.99% to 14.36% in G0 and from 4.06% to 15.53% in G1, relative to each family's DDIM/Complex-L1 baseline. Its mean AUCs are 0.8076 and 0.8075. Within-family module comparisons, full rotation curves, and frozen-threshold results support this verification design. Identification is evaluated separately and remains weak under rotation. Component measurements are available; integrated task-routing results remain pending.

1. Introduction

Provenance information supports attribution and tracing as generated images circulate, and must remain verifiable after editing. Generative watermarking embeds a source signal during synthesis for subsequent testing against a registered key [5, 10]. A missed watermark breaks this link, while a false positive associates an unrelated image with a source. The detector must therefore recover watermarks from processed images while keeping false positives low. Geometric transformations make this requirement particularly demanding because they change the spatial correspondence needed for watermark recovery.

Frequency-domain watermarking embeds structured patterns in the initial diffusion noise and recovers them by inverting the generated image. Tree-Ring established this approach [10]. SFWMark subsequently preserved Fourier integrity through Hermitian-symmetric embedding and improved crop robustness through center-aware placement [5]. These advances strengthen the embedded representation, but reliable detection also depends on how the detector recovers and compares it. In SFWMark's HSQR variant, the image is encoded, inverted with DDIM, and matched to a key template using Complex-L1.

Rotation exposes difficulties at both stages. On a finite canvas, it changes frequency coordinates and introduces interpolation errors and boundary losses. DDIM inversion adds approximation error when recovering the initial latent [3, 9], so the recovered coefficients can deviate from the key even when watermark-related structure remains. The resulting residual combines geometric distortion with encoding and inversion effects. Complex-L1 weights coefficient discrepancies equally [5]; it does not account for a systematic residual offset or for directions in which errors are more variable or correlated. Improving inverse-step consistency therefore leaves a second problem: how to interpret the residual that remains.

The decision objective further determines what a useful residual score should measure. Verification asks whether an image contains a specified key, whereas identification ranks a set of candidate keys. The original SFWMark detector uses Complex-L1 for both tasks [5]. Yet compatibility with the residual distribution of a correct key and relative similarity across competing templates are different comparisons. A metric that improves verification need not improve identification. These distinctions motivate a detector that combines more consistent recovery with statistical residual modeling and a decision rule matched to the requested task.

We propose TAIR-WM, a training-free detector that implements this approach while retaining HSQR embedding. Neural-network weights remain fixed, residual statistics are fitted offline, and the caller specifies verification or identification. Our three technical contributions address the corresponding difficulties:

1. **Residual-Guided Exact-DDIM Inversion.** We adapt the forward-step refinement principle of Exact-DPM [3]

to HSQR recovery. Starting from DDIM, we refine candidates according to their local reconstruction residuals, with a finite evaluation budget and a residual-based stopping rule. Retaining each image's best evaluated candidate bounds its local residual by that of the initialization.

2. **Covariance-Aware Watermark Verification.** We model correct-key feature residuals using their mean and a Ledoit–Wolf shrinkage covariance [4]. The resulting Mahalanobis score evaluates residual compatibility after accounting for systematic offsets and correlated variation.
3. **Task-Aware Dual-Metric Detection.** We define task-specific decisions over shared recovered features: Mahalanobis scoring for specified-key verification and Complex-L1 matching for candidate-key identification. This design allows the verification branch to use fitted residual statistics while preserving direct template ranking for identification.

Experiments show improved verification under rotation. Averaged over twelve nonzero bilinear rotation conditions, the Exact-DDIM/Mahalanobis combination raises TPR at FPR below 1% from 3.99% to 14.36% in configuration family G0 and from 4.06% to 15.53% in G1, compared with each family's DDIM/Complex-L1 baseline. These are gains of 10.37 and 11.47 percentage points under the same false-positive constraint. Module comparisons show that the benefit of inverse refinement depends on the scoring rule, supporting the joint design of recovery and verification. Separate identification measurements favor Complex-L1 over Mahalanobis on the original twelve-condition suite, although identification remains weak under rotation. The task-specific components have been evaluated separately; their integrated routing evaluation remains pending.

2. Related Work

Generative frequency-domain watermarks. Latent diffusion permits watermark insertion before decoding [6]. Tree-Ring embeds Fourier structures in initial noise and detects them through inversion [10]. SFWMark uses Hermitian symmetry and central embedding to retain frequency integrity [5]. TAIR-WM keeps HSQR embedding and feature extraction, adapting recovery and decision scoring.

Task objectives and geometric robustness. RingID studies binary detection, multi-key identification, and discrete Fourier patterns under rotation [1]. SynTag combines synchronization features, learned prediction, and geometric correction [2]. The original SFWMark already distinguishes verification from identification, using Complex-L1 for both [5]. Our task distinction is therefore not itself new. We assign different residual metrics and output rules to these tasks, without adding angle estimation or registration.

Diffusion inversion. DDIM provides deterministic sampling and a practical inverse [9]. Exact-DPM treats inverse steps as implicit problems and studies forward-step and gradient-based refinement, including watermark applications [3]. GNRI uses guided Newton–Raphson iterations [7]; FARI studies robust one-step inversion with adversarial adaptation [11]. We adapt the forward-step mechanism to first-order DDIM in HSQR detection. The residual update is inherited; finite-budget selection, residual scoring, and task-specific decisions define the present detector configuration.

Statistical scoring and auxiliary feature modulation. Ledoit–Wolf shrinkage stabilizes high-dimensional covariance estimation by combining a sample covariance with a scaled identity [4]. We apply this established estimator to correct-key watermark residuals. FreeU rescales backbone features and modulates the low-frequency components of skip features inside a diffusion U-Net [8]. Our code invokes existing FreeU functionality during both generation and inversion in G1. It introduces no watermark-conditioned FreeU mechanism; FreeU remains an auxiliary configuration rather than a separate contribution.

3. Method

3.1. Problem Formulation and Overview

Let x be an observed image and $\mathcal{K}$ a set of registered keys. Verification tests a caller-specified key k; identification selects among supplied candidates. The task is $q \in \{\mathrm{V}, \mathrm{I}\}$. HSQR embedding is retained. A VAE encoder E, diffusion inverse I, and Fourier feature map Φ share the recovery path:

$$z_0 = E(x), \quad \widehat{z}_T = I(z_0), \quad f = \Phi(\widehat{z}_T). \tag{1}$$

Here E includes the base model's latent scaling and $\widehat{z}_T$ is the recovered initial-noise representation. Figure 1 shows the three technical stages. Recovery can be reused if both tasks are requested.

For the center-aware configuration, Φ crops a 44×44 latent region, applies a shifted two-dimensional FFT, and selects 42×21 complex coefficients from channel index 3. Let $c \in \mathbb{C}^m$ flatten these coefficients, with $m = 882$. Then

$$f = [\mathrm{Re}(c)^\top, \mathrm{Im}(c)^\top]^\top \in \mathbb{R}^p, \quad p = 2m = 1764. \tag{2}$$

Template w_k follows the same ordering. Its real and imaginary parts use the two halves of a QR pattern with reference values ± 45, as in the inherited detector. Given $r_k = f - w_k$, verification measures compatibility with a fitted correct-key residual distribution; identification compares template distances across $\mathcal{K}$.

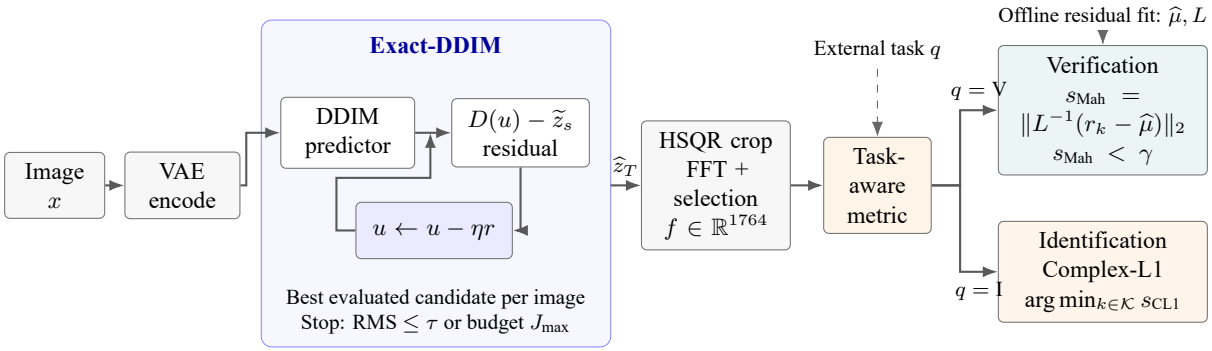


Figure 1. TAIR-WM detection framework. Residual-guided Exact-DDIM supplies the shared HSQR features; covariance-aware verification scores a specified key; the external task selects this branch or Complex-L1 identification. Dashed inputs are offline statistics or caller-supplied control, and solid arrows carry inference data. The three contributions concern recovery, verification modeling, and task-dependent decisions, respectively. No learned router is introduced.

3.2. Residual-Guided Exact-DDIM Inversion

Local consistency objective. For adjacent levels $t > s$, let $D_{t\to s}$ be a deterministic DDIM denoising step. In noise-prediction notation,

$$\begin{aligned} \widehat{z}_0(z_t) &= \frac{z_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon_\theta(z_t, t)}{\sqrt{\bar{\alpha}_t}}, \\ D_{t\to s}(z_t) &= \sqrt{\bar{\alpha}_s}\,\widehat{z}_0(z_t) + \sqrt{1-\bar{\alpha}_s}\,\epsilon_\theta(z_t, t). \end{aligned} \tag{3}$$

The implementation follows the base scheduler's coefficients and parameterization. Because the prediction depends on the unknown z_t, inversion is an implicit problem. A DDIM inverse approximates its solution but need not reconstruct the current lower-noise target $\widetilde{z}_s$. We measure this discrepancy by

$$\mathcal{L}_t(z) = \frac{1}{d}\|D_{t\to s}(z) - \widetilde{z}_s\|_2^2. \tag{4}$$

where d is the number of latent entries per image. The target is fixed within the refinement loop, and ground-truth initial noise is unnecessary.

Residual correction. Initialize with the ordinary inverse predictor and evaluate the denoised reconstruction:

$$z_t^{(0)} = I_{s\to t}^{\mathrm{DDIM}}(\widetilde{z}_s), \quad r_t^{(j)} = D_{t\to s}(z_t^{(j)}) - \widetilde{z}_s. \tag{5}$$

The forward-step correction is

$$z_t^{(j+1)} = z_t^{(j)} - \eta r_t^{(j)}. \tag{6}$$

This is the Exact-DPM forward-step mechanism [3], adapted to the first-order DDIM detector. It subtracts the latent reconstruction residual without differentiating through the denoiser. It is not gradient descent on Eq. (4), which would require the denoising map's Jacobian.

We use $\eta = 0.5$, a maximum of $J_{\max} = 5$ candidate evaluations per step, and tolerance $\tau = 10^{-5}$. The scheduler's stochastic noise parameter is zero. The selected candidate becomes the target of the next inverse step.

Stopping and per-image retention. Refinement stops when the maximum residual RMS across the batch is at most τ, or when the budget is exhausted. The maximum criterion requires every image to meet the tolerance, rather than allowing a small batch average to conceal a large residual.

Updates need not improve monotonically. For each image b, retain the candidate with minimum evaluated MSE:

$$j_b^\star = \arg\min_{j \in \mathcal{J}_b} \mathcal{L}_t(z_{t,b}^{(j)}), \quad \widetilde{z}_{t,b} = z_{t,b}^{(j_b^\star)}. \tag{7}$$

The evaluated set $\mathcal{J}_b$ includes initialization. An unevaluated final update is ineligible. For finite evaluated losses, this immediately gives

$$\mathcal{L}_t(\widetilde{z}_{t,b}) \leq \mathcal{L}_t(z_{t,b}^{(0)}). \tag{8}$$

This property concerns the fixed target of one step. It guarantees neither strict improvement nor global convergence, and does not recover content lost through rotation. The term Exact-DDIM denotes the adopted refinement family, not exact recovery under a finite budget. Algorithm 1 specifies candidate evaluation and task decisions.

3.3. Covariance-Aware Watermark Verification

Correct-key residual model. For fitting features f_i with known correct templates w_{k_i}, set $r_i = f_i - w_{k_i}$. Their mean captures systematic displacement; covariance captures variation and correlations between Fourier locations and real/imaginary components:

$$\widehat{\mu} = \frac{1}{n}\sum_{i=1}^{n} r_i, \quad S = \frac{1}{n}\sum_{i=1}^{n}(r_i - \widehat{\mu})(r_i - \widehat{\mu})^\top. \tag{9}$$

The recovered distribution need not be centered on the template. Likewise, geometric disturbances can move several coefficients together, making equally scaled independent discrepancies an incomplete residual model.

With $n = 1024$ fitting samples and $p = 1764$ dimensions, the sample covariance is rank deficient. We use Ledoit–Wolf shrinkage [4]:

$$\widehat{\Sigma} = (1-\lambda)S + \lambda a I, \quad a = \frac{\mathrm{tr}(S)}{p}, \qquad \Sigma_{\mathrm{reg}} = \widehat{\Sigma} + \varepsilon I. \tag{10}$$

The estimator determines λ from the fitting data. Nonnegative jitter ε is added only if Cholesky factorization requires it. These are standard estimation tools; our modeling choice is to fit them to the detector's correct-key residuals, separately for each inversion configuration.

Whitened compatibility score. Factor $\Sigma_{\mathrm{reg}} = LL^\top$ and solve $Lv_k = r_k - \widehat{\mu}$. Define

$$s_{\mathrm{Mah}}(r_k) = \|v_k\|_2 = \sqrt{(r_k - \widehat{\mu})^\top \Sigma_{\mathrm{reg}}^{-1}(r_k - \widehat{\mu})}. \tag{11}$$

The implementation uses a triangular solve instead of forming the matrix inverse. Higher-variance directions receive lower weight, and off-diagonal terms represent correlated deviations. This score measures residual compatibility with the fitted correct-key population. We do not assume a complete Gaussian likelihood model for both watermarked and unwatermarked images.

Covariance weighting can accommodate recurring disturbances represented in fitting data, but its value depends on that distribution. Full-rank whitening is invertible; it does not inherently delete high-frequency coefficients. The following norm compresses the vector to a scalar. Thus a possible reduction in key discriminability concerns directional weighting, not a proved removal of particular frequency bands. This motivates separating verification from key ranking.

3.4. Task-Aware Dual-Metric Detection

Verification compares one specified key against an unwatermarked alternative; identification ranks many candidates. Our task-aware dual-metric rule, also termed the Task-Aware Adaptive Metric, is

$$s_q(f, w_k) = \begin{cases} s_{\mathrm{Mah}}(f - w_k), & q = \mathrm{V}, \\ s_{\mathrm{CL1}}(f, w_k), & q = \mathrm{I}. \end{cases} \tag{12}$$

For complex coefficients c_j and template coefficients $u_{k,j}$, the identification distance is

$$s_{\mathrm{CL1}}(f, w_k) = \frac{1}{m}\sum_{j=1}^{m} |c_j - u_{k,j}|, \qquad |a + ib| = \sqrt{a^2 + b^2}. \tag{13}$$

Algorithm 1 Exact-DDIM with task-aware dual-metric scoring

Require: Image x, task q, key or key set, fitted $(\widehat{\mu}, L)$, threshold γ

$z \leftarrow E(x)$
for each inverse step (s, t) in increasing noise **do**
  $\widetilde{z}_s \leftarrow z$; $u \leftarrow I^{\mathrm{DDIM}}_{s\to t}(\widetilde{z}_s)$
  $e_b^\star \leftarrow +\infty$ for every image b
  for $j = 0, \ldots, J_{\max} - 1$ **do**
    $r \leftarrow D_{t\to s}(u) - \widetilde{z}_s$
    $e_b \leftarrow \mathrm{mean}(r_b^2)$ for each b
    if $e_b < e_b^\star$ for an image b **then**
      $u_b^\star \leftarrow u_b$; $e_b^\star \leftarrow e_b$
    end if
    if $\max_b \sqrt{e_b} \le \tau$ **then**
      break
    end if
    $u \leftarrow u - \eta r$
  end for
  $z \leftarrow u^\star$
end for
$f \leftarrow \Phi(z)$
if $q = \mathrm{V}$ **then**
  $v \leftarrow L^{-1}(f - w_k - \widehat{\mu})$
  return $\mathbf{1}[\|v\|_2 < \gamma]$
else
  return $\arg\min_{k\in\mathcal{K}} s_{\mathrm{CL1}}(f, w_k)$
end if

Complex-L1 is the mean complex modulus, not the sum of separate absolute real and imaginary differences. It preserves the base detector's equal weighting across complex locations. The task outputs are

$$\widehat{y} = \mathbf{1}[s_{\mathrm{Mah}}(r_k) < \gamma] \quad (q = \mathrm{V}), \qquad \widehat{k} = \arg\min_{k\in\mathcal{K}} s_{\mathrm{CL1}}(f, w_k) \quad (q = \mathrm{I}). \tag{14}$$

Threshold γ is calibrated independently of the test image. Identification is closed-set selection and does not use γ to reject an unknown key.

The new interface separates statistical acceptance of a specified key from direct ranking of alternatives while sharing f. Task selection is externally controlled. It uses no learned gate, attack classifier, or per-image metric search. Both score branches have component measurements; their integrated execution remains [TBD]. The prescribed identification branch is not claimed to outperform every other metric under all attacks.

3.5. Implementation and Computational Cost

We use Stable Diffusion 2.1 base at 512×512. Images are resized, mapped to $[-1, 1]$, and encoded with the VAE posterior mode and latent scaling. Inversion uses 50 steps, fp32, an empty prompt, and guidance scale zero; neural weights remain fixed.

Each inverse step needs one predictor and up to five denoising evaluations:

$$N_{\text{eval}} = T + \sum_{t=1}^{T} J_t \le T(1 + J_{\max}), \qquad T = 50, \quad J_{\max} = 5, \quad N_{\text{eval}} \le 300. \tag{15}$$

This is an evaluation budget, not a measured latency ratio. Candidate retention is online and need not store the trajectory. Offline fitting and factorization require no additional neural training. Full-covariance verification uses an $O(p^2)$ triangular solve and $O(p^2)$ stored statistics. Complex-L1 identification costs $O(Km)$ for K keys and permits chunked templates. The task branch adds only a conditional selection.

Statistics and thresholds are configuration-specific. Diagonal scoring uses the same fitted mean and the diagonal of the shrunk covariance with a numerical floor. The frozen threshold is the 20th ordered clean negative distance among 2000 calibration negatives, with the strict comparison in Eq. (14).

G0 disables FreeU; G1 enables the existing pipeline function in both generation and inversion, with $(s_1, s_2, b_1, b_2) = (0.9, 0.2, 1.4, 1.6)$. It modifies the U-Net used by the predictor and denoising evaluations, but adds no key-conditioned module or change to HSQR embedding. Its internal feature-frequency modulation is distinct from directly editing the watermark's Fourier template.

4. Experiments

We evaluate the three technical stages through module combinations and task comparisons. The former assess recovery and verification scoring; the latter assess the rationale and limits of the two decision rules. Full rotation curves and frozen thresholds expose failures hidden by averages. Auxiliary configurations contextualize these results.

4.1. Experimental Setup

We use the existing Q0–Q5 records. Q0/Q1 share G0 and Q3/Q4 share G1; each pair replaces DDIM with Exact-DDIM. Q2/Q5 use GNRI in the corresponding families. Within-family comparisons retain the generation pool and FreeU setting. Across families, both the pool and FreeU status change, precluding isolated FreeU attribution.

The split sizes are 1024 fitting, 2000 calibration, 400 development samples, and 1000 test pairs. Each condition contains 1000 watermarked and 1000 corresponding unwatermarked images. Fitting uses eight groups of 128: clean, JPEG, noise, center crop, random crop, and absolute rotations of 5°, 15°, and 30°, each rotation group including both signs. Evaluation therefore includes fitting-covered and additional angles.

The main test is fixed-canvas bilinear rotation at

$$\theta \in \{\pm 5, \pm 10, \pm 15, \pm 30, \pm 45, \pm 75\}^\circ.$$

Rotation means weight all twelve nonzero angles equally. Zero rotation is a separate control; nearest-neighbor and orthogonal rotations are excluded. Curves retain signed angles because the two directions need not behave identically.

We report AUC and maximum TPR among test-ROC points satisfying FPR $< 1\%$, using negative distance as the positive-class score. With 1000 negatives, at most nine false positives are eligible. This is distinct from a preselected threshold: Frozen TPR/FPR instead apply the clean-calibrated threshold. AUCs and rates are averaged per condition, not pooled across scores. These are point estimates; no confidence intervals or significance claims are supplied.

Identification accuracy measures correct selection among 2048 keys. The Original-12 aggregate contains clean, brightness, contrast, JPEG, blur, noise, BM3D, two VAE conditions, diffusion regeneration, center crop, and random crop. All twelve source conditions are retained for task comparison; rotation remains the primary verification setting.

4.2. Main Results under Rotation

Table 1 and Figure 2 report the main results. Exact-DDIM/Mahalanobis reaches mean AUC 0.8076 and TPR 14.36% in G0, and AUC 0.8075 and TPR 15.53% in G1. Against each family's DDIM/Complex-L1 baseline, TPR increases by 10.37 and 11.47 percentage points, calculated from displayed values. These gains concern the combined verification branch, not inversion alone.

Under Mahalanobis, changing DDIM to Exact-DDIM raises G1 AUC at all twelve angles. TPR rises at ten, is unchanged at -45°, and falls from 11.8% to 10.8% at $+45^\circ$. In G0, AUC and TPR each increase at eleven angles, with TPR decreasing slightly at -75°. Better full-ROC ranking need not improve every operating point.

Large angles remain difficult. Combined-branch TPR is 4.5%/4.9% in G0/G1 at $+75^\circ$, and 4.4%/3.7% at -75°. The method changes numerical recovery and residual scoring without aligning the rotated template.

With Mahalanobis, G0 Frozen TPR rises from 4.50% to 8.60%, while Frozen FPR changes from 0.25% to 0.35%. G1 changes from 5.38% to 9.49% TPR and 0.31% to 0.38% FPR. These results show gains with thresholds fixed before attack evaluation. Nevertheless, G1 Exact-DDIM/Mahalanobis has Frozen FPR 1.2% at $+5^\circ$. Clean calibration does not guarantee the nominal target for every attacked condition.

4.3. Recovery and Scoring Ablation

Retaining Complex-L1 while replacing DDIM with Exact-DDIM changes mean TPR from 3.99% to 3.95% in G0 and

Table 1. Means over all twelve nonzero bilinear rotations. AUC is unitless; other entries are percentages. The four rows per family cross two inversions with two scores. Frozen rates use configuration-specific clean-calibrated thresholds. Bold identifies the combined verification configuration, not the best value in every column.

Family	Inversion	Score	AUC	TPR (< 1%)	Frozen TPR	Frozen FPR
G0	DDIM	Complex-L1	0.6305	3.99	2.49	0.60
G0	Exact-DDIM	Complex-L1	0.6274	3.95	2.52	0.56
G0	DDIM	Mahalanobis	0.7892	10.20	4.50	0.25
G0	Exact-DDIM	Mahalanobis	**0.8076**	**14.36**	**8.60**	**0.35**
G1	DDIM	Complex-L1	0.6299	4.06	2.69	0.50
G1	Exact-DDIM	Complex-L1	0.6279	4.32	2.33	0.52
G1	DDIM	Mahalanobis	0.7878	10.48	5.38	0.31
G1	Exact-DDIM	Mahalanobis	**0.8075**	**15.53**	**9.49**	**0.38**

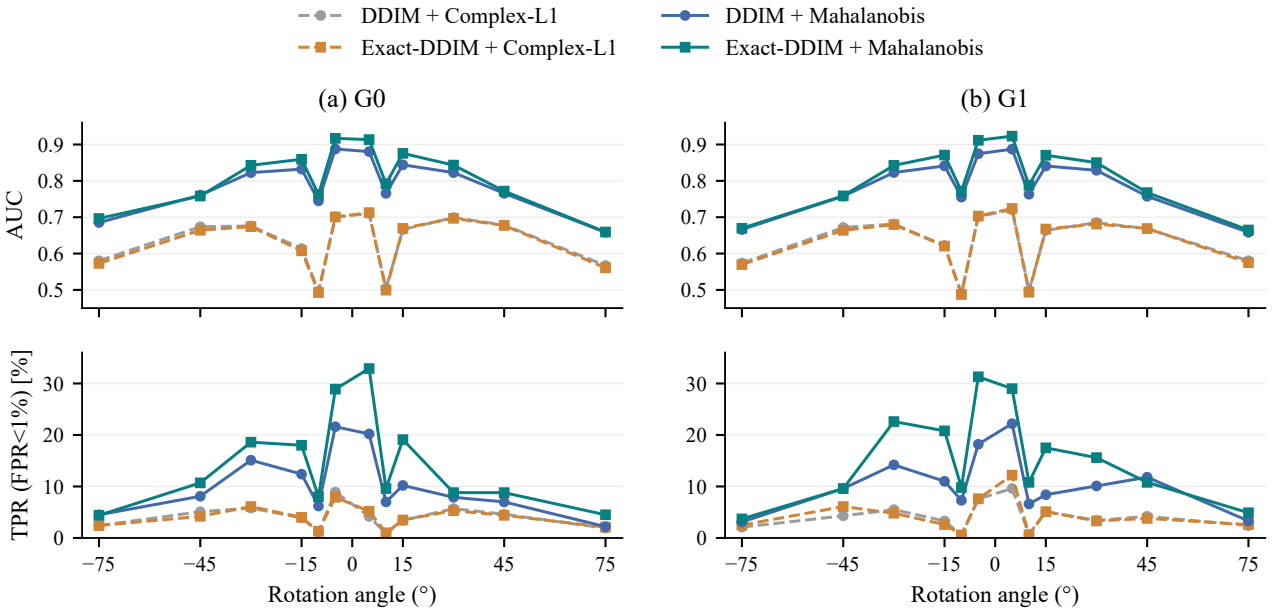


Figure 2. Verification at all twelve nonzero bilinear angles. Top: AUC; bottom: TPR at FPR below 1%. Columns show G0 and G1. Every point is an existing per-angle measurement. Lines connect measured angles for readability; no uncertainty bands or intermediate measurements are inferred. The complete range, including large-angle failures, is retained.

4.06% to 4.32% in G1. AUC changes from 0.6305 to 0.6274 and 0.6299 to 0.6279. Refinement alone with the baseline score does not reproduce the combined improvement.

Under DDIM, replacing Complex-L1 with Mahalanobis raises TPR to 10.20% in G0 and 10.48% in G1. Exact-DDIM then adds 4.16 and 5.05 percentage points. These contrasts support empirical complementarity between the recovery and scoring choices. Define

$$\Delta_{\text{int}} = (M_{\text{Exact,Mah}} - M_{\text{DDIM,Mah}}) - (M_{\text{Exact,CL1}} - M_{\text{DDIM,CL1}}). \tag{16}$$

where M is mean rotation TPR. Displayed values give 4.20 pp in G0 and 4.79 pp in G1. This descriptive interaction is not a causal decomposition or significance test: statistics are refitted per inversion, and the score replacement changes both centering and covariance weighting.

For Exact-DDIM, L2 and diagonal scoring give TPR 5.03%/7.52% in G0, versus 14.36% for full Mahalanobis; G1 gives 5.05%/8.26%, versus 15.53%. Diagonal and full scores share the fitted mean and covariance estimate, so their contrast assesses off-diagonal structure within this implementation. L2-to-diagonal changes include centering and directional scaling and are not a pure correlation ablation.

4.4. Task-Aware Scoring

Table 2 separates the two task objectives. For Exact-DDIM, Mahalanobis gives stronger rotation verification than Complex-L1. On Original-12, Complex-L1 identifica-

Table 2. Task-specific scores with Exact-DDIM (G0/Q1 and G1/Q4). Entries are percentages; identification uses 2048 keys. CL1 and Mah denote Complex-L1 and Mahalanobis. Integrated routing remains pending.

Condition / metric	Score	G0	G1
Rotation / TPR	CL1	3.95	4.32
Rotation / TPR	Mah	14.36	15.53
Original-12 / ID	CL1	98.73	98.73
Original-12 / ID	L2	98.80	98.78
Original-12 / ID	Mah	80.46	80.45
Rotation / ID	CL1	0.13	0.13
Rotation / ID	Mah	0.09	0.08
Router / rotation TPR	Mah	[TBD]	[TBD]
Router / Original-12 ID	CL1	[TBD]	[TBD]

tion is 98.73% in both families, compared with Mahalanobis at 80.46%/80.45%. These component results motivate different scores for specified-key compatibility and multi-key ranking.

Complex-L1 is not universally optimal. L2 identification on Original-12 is slightly higher, at 98.80%/98.78%. We retain the base detector's complex-modulus rule as an explicit identification branch, without choosing a different metric for every attack.

Rotation identification remains weak: Complex-L1 averages 0.13% in both families at source precision, while Mahalanobis yields 0.09%/0.08%. Original-12 accuracy cannot be transferred to rotated images. Verification gains therefore do not establish recovery of multi-key identity.

The final rows reserve integrated routing evaluation, which must check whether the interface reproduces each separate branch under the same requests and candidate sets. Each task should retain its own metric, rather than a mixed accuracy. Results remain [TBD]; the component averages are not measurements of the integrated router.

4.5. Auxiliary Configurations

GNRI [7] with Mahalanobis reaches AUC 0.7986 and TPR 13.11% in G0, and 0.7978/13.72% in G1, below Exact-DDIM on these means. However, G0 GNRI Frozen TPR is 9.23%, above 8.60%, with Frozen FPR 0.46% rather than 0.35%. Conclusions depend on the endpoint. GNRI uses two internal iterations, versus up to five Exact-DDIM candidate evaluations, so this is not an equal-computation comparison. Matched-device end-to-end latency, including both score branches, remains [TBD].

FreeU changes the U-Net feature flow in both G1 generation and inversion. It can consequently change the generated signal and recovered residual distribution; a specific watermark-frequency effect is not established. Separate statistical fitting can accommodate configuration-dependent residuals, but does not constitute a new FreeU collaboration mechanism. Since generation pools and both FreeU stages change together, the G0/G1 contrast isolates neither a FreeU effect nor its interaction with Exact-DDIM or Mahalanobis. The present formal records do not supply the matched, phase-specific evidence needed for those claims.

5. Discussion

Distinct technical roles. TAIR-WM addresses local inverse consistency, correct-key residual compatibility, and task-dependent decisions. Best-candidate selection gives a local non-increase property; covariance scoring uses an empirical residual distribution; the task interface assigns different scores to different outputs. Neither the residual update nor Mahalanobis mathematics is claimed as newly invented. Their HSQR adaptation and division of detector responsibilities define the contribution.

Evidence and limits. The complete rotation evaluation supports the combined verification branch, but does not identify which frequency directions mediate its gains. Low large-angle TPR and rotated identification show that synchronization remains unresolved. Fitting uses a specified mixture of disturbances and does not establish superiority for every attack distribution. Brightness and noise are not separately analyzed here.

Integrated routing, matched-device latency, and isolated FreeU mechanism tests remain unverified. Unknown-key search also needs a distinct false-positive analysis because testing multiple keys changes the decision event. These limits constrain the interpretation of the present component measurements.

6. Conclusion

TAIR-WM retains HSQR embedding and fixed neural weights while separating three detector functions: residual-guided Exact-DDIM inversion, covariance-aware watermark verification, and task-aware dual-metric detection. Finite-budget candidate retention guarantees no worse evaluated local consistency than initialization. Correct-key residual whitening improves the combined branch's mean verification AUC and low-FPR TPR on the twelve-angle grid in both families. Complex-L1 supplies a separately evaluated identification branch. Full module comparisons, frozen thresholds, and task-specific results preserve the empirical scope: rotation identification remains weak, and integrated routing is still to be measured.

References

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